CHAPTER

70

Spleen

The spleen is a large, encapsulated, complex mass of vascular and lymphoid tissue situated in the upper left quadrant of the abdominal cavity between the fundus of the stomach and the diaphragm. It is mainly concerned with phagocytosis and immune responses. Although documented for more than 3,000 years, there are many aspects of the spleen that remain poorly understood. It plays important roles in immunological defence, metabolism and maintenance of circulating blood elements (Petroianu 2011, Tarantino et al 2013; **Table 70.1**). In the fetus, it is also a major site of haemopoiesis and can resume this role postnatally in certain pathological conditions (Üngör et al 2007). However, it is not essential for life; if the spleen is removed, many of its functions can be assumed by the liver and other tissues of the mononuclear phagocytic system.

The most common clinical manifestations of splenic disorders are splenomegaly and a decrease in the number of cellular elements in the blood (cytopenias). Other clinical manifestations include infections, lassitude and abdominal discomfort (Coetzee 1982, Oguro et al 1993, Petroianu 2011, Tarantino et al 2013). Chronic splenomegaly in children may be associated with growth retardation (Petroianu 2003, Petroianu 1996). The most serious long-term consequence of removal of the spleen is severe sepsis, which carries a 2% mortality in otherwise healthy adults and an even greater risk of death in children, the elderly, and patients with chronic diseases.

GROSS ANATOMY

The size and weight of the spleen vary with age and sex, and can also vary slightly in the same individual under different conditions. The adult spleen is usually 9–14 cm long, 6–8 cm wide and 3–5 cm thick, and fits comfortably in the individual's cupped hand. It reaches its largest dimension in puberty and diminishes thereafter (DeLand 1970, Jakobsen and Jakobsen 1997). Although its weight increases during puberty, it is comparatively largest in the young child and tends to diminish in size and weight in old age (Coquet et al 2010, DeLand 1970, Jakobsen and Jakobsen 1997, Scothorne 1985, Skandalakis et al 1993, Üngör et al 2007). The average adult weight is dependent on the volume of contained blood; emptied of blood, it weighs between 70 and 120 g, whereas *in vivo* its weight ranges from 150 to 350 g (Nakamura et al 1989, Petroianu 2011, Skandalakis et al 1993).

The shape of the spleen is also variable and mostly determined by its relations to neighbouring structures during development; it often appears as a slightly curved wedge. The superolateral aspect is shaped by the left dome of the diaphragm, and the inferomedial aspect mostly by the neighbouring stomach, left kidney and splenic flexure of the colon. In the fetus, the spleen is lobulated (Ch. 60; Coetzee 1982, Faller 1985, Gupta et al 1976, Petroianu 2011, Scothorne 1985, Skandalakis et al 1993, Üngör et al 2007); the adult spleen usually only retains a notch on its anterior border, although additional surface notches may persist (Mikhail et al 1979).

A splenic lobule that fails to coalesce with the developing spleen can persist as a supernumerary or accessory spleen (also known as a splenunculus; Gupta et al 1976). This fully functional island of splenic tissue is found in approximately 10% of individuals and may be located in any part of the abdomen, or even outside it, but is most commonly present near the splenic hilum within the gastrosplenic ligament or greater omentum (Faller 1985, Petroianu 2011; **Fig. 70.1**).

A normal adult spleen is not palpable on abdominal examination. In living supine healthy adults, it is most frequently located between the tenth and twelfth ribs, with its long axis along the eleventh rib (Gupta et al 1976, Mirjalili et al 2012). Its posterior border is approximately 4 cm from the midline at the level of the tenth thoracic vertebral spine and it extends about 3 cm anterior to the mid-axillary line. In the absence of long peritoneal ligaments, it has to triple in size before it becomes palpable below the left costal margin (Mikhail et al 1979, Üngör et al 2007).

RELATIONS

The spleen has superolateral diaphragmatic and inferomedial visceral surfaces (**Fig. 70.2**), anterosuperior and posteroinferior borders, and

Table 70.1 Summary of splenic functions

General functions	Specific functions
Haematological and immunological	Haemopoiesis Maturation of blood elements Immunoglobulin activation Recirculation of T and B lymphocytes
Production	Leukocytes Peptides Immunoglobulins (mainly IgM) Opsonins Tuftsin Properdin Complement factors Stem cells
Storage	Leukocytes Platelets All metals
Clearance	Microorganisms Parasites Circulating antigens Altered circulating cells Circulating foreign bodies
Synthesis	Precursor of hepatic functions
Metabolism	Lipids Cholesterol Bilirubin Amino acids
Control	Bone marrow Mononuclear phagocytic function Somatic growth

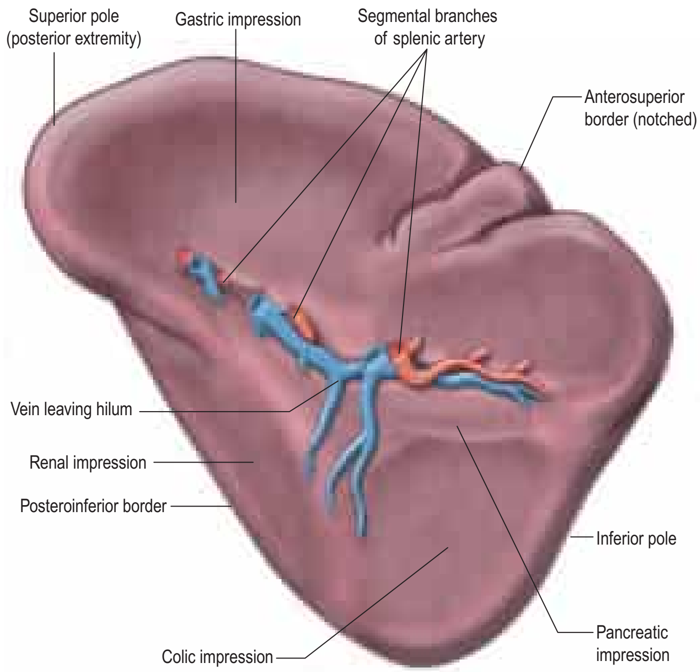

Fig. 70.2 The visceral surface of the spleen.

Spleens in men are proportionally heavier than those in women. In pathological conditions, the spleen may expand and retain blood; weights rarely exceed 700 g but gigantic spleens exceeding 10 kg have been reported (Coquet et al 2010, DeLand 1970, Jakobsen and Jakobsen 1997, Skandalakis et al 1993). Splenomegaly compresses and displaces adjacent organs, causing abdominal discomfort, dyspepsia, respiratory restriction and difficulty walking (Coetzee 1982, Petroianu 2011).

Fig. 70.1 A supernumerary spleen located in the greater omentum (arrow). Supernumerary spleens are usually isolated and may be connected to the spleen or splenic pedicle by thin vessels.

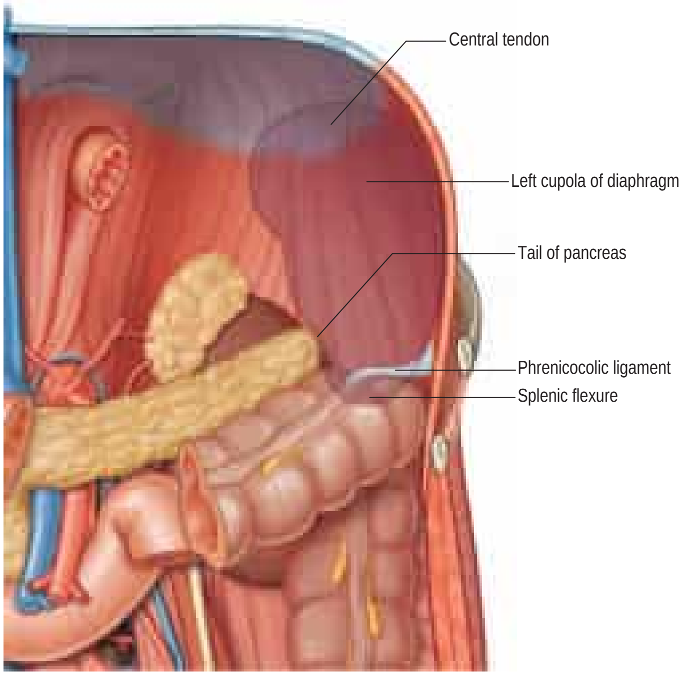

Fig. 70.3 The posterior relations ('bed') of the spleen. (Adapted from Drake, RL, Vogl, AW, Mitchell, A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

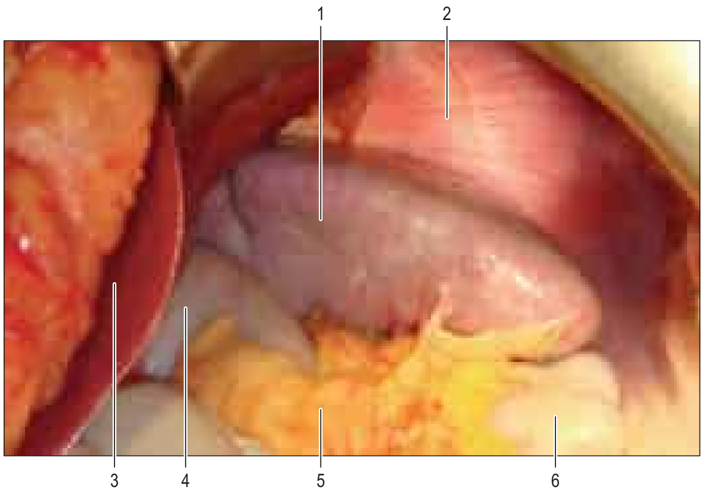

Fig. 70.4 The gross appearance of the spleen *in situ*. An intra-abdominal image of the spleen (1) and its relation with the stomach (4), liver (3), colon (6), diaphragm (2) and greater omentum (5).

superior and inferior poles. The convex, smooth diaphragmatic surface faces mostly superiorly and laterally, although the posterior part may face posteriorly (Mirjalili et al 2012). The diaphragmatic surface is separated from the left pleural costodiaphragmatic recess, lower lobe of the left lung and the tenth to twelfth left ribs by the underside of the left dome of the diaphragm (**Figs 70.3**–**70.4**); splenic inflammation or surgery may lead to a left-sided basal pleural effusion and left lower lobe atelectasis (Petroianu 2011).

The visceral surface is irregular, faces inferomedially towards the abdominal cavity and is marked by gastric, renal and colic impressions. The gastric impression faces anteromedially and is broad and concave where the spleen lies adjacent to the posterior aspect of the fundus, upper body and upper greater curvature of the stomach. It is separated from the stomach by a peritoneal recess, limited by the gastrosplenic ligament. The renal impression is slightly concave and lies on the posteroinferior part of the visceral surface, separated from the gastric impression above by a ridge of splenic tissue and the splenic hilum. It faces inferomedially and slightly backwards, and is related to the upper lateral area of the anterior surface of the left kidney and sometimes to the superior pole of the left suprarenal gland. The colic impression is usually flat; it lies at the inferior pole of the spleen and is related to the splenic flexure of the colon and the phrenicocolic ligament. The hilum of the spleen is a long fissure pierced by the splenic vessels, nerves and lymphatics, and lies on the visceral surface closer to the posteroinferior border (Petroianu 2011).

The anterosuperior border separates the diaphragmatic surface from the gastric impression and is usually convex. Inferiorly, it may bear one or two notches that have persisted from the lobulated form of the spleen in early fetal life (p. 1064). However, the notch may be absent and is not a completely reliable guide to identification of the spleen during clinical examination. The posteroinferior border separates the renal impression from the diaphragmatic surface and is more rounded and blunt than the anterosuperior border. The superior pole corresponds to the posterior extremity and usually faces the vertebral column. The inferior pole is longer and less angulated than the superior pole and connects the anterosuperior and posteroinferior borders anteriorly; it is related to the colic impression and often lies adjacent to the splenic flexure and phrenicocolic ligament (Petroianu 2011).

The inferior pole of the spleen is particularly at risk of injury from blunt abdominal trauma or during surgical procedures on the stomach, pancreatic tail, left kidney, left suprarenal gland and left colon. Excessive traction of the stomach, transverse colon or greater omentum may tear the splenic capsule and superficial parenchyma by way of their peritoneal attachments, causing bleeding that may be difficult to control (Merchea et al 2012).

SPLENIC LIGAMENTS

The spleen develops between the two leaves of the dorsal mesogastrium (see Figs 60.6–60.7) and so is almost entirely invested in visceral peritoneum that is firmly adherent to its capsule (Üngör et al 2007). The dorsal mesogastric attachments persist as peritoneal ligaments. Thus, the superior pole of the spleen is connected to the stomach via the gastrosplenic ligament, and to the posterior abdominal wall by a variably developed phrenicosplenic ligament. The inferior pole of the spleen is connected to the posterior abdominal wall by the splenorenal ligament and to the splenic flexure of the colon. The phrenicocolic ligament lies just inferior to the lower pole (**Fig. 70.5**). Each of these ligaments is made up of two layers of peritoneum containing fat, blood and lymphatic vessels and nerves (Petroianu 2011, Skandalakis et al 1993). The phrenicosplenic ligament runs between the spleen and the peritoneum of the undersurface of the diaphragm. The anterior layer of the splenorenal ligament is continuous with the peritoneum of the posterior wall of the lesser sac over the left kidney, and with the posterior layer of the gastrosplenic ligament at the splenic hilum. The posterior layer of the splenorenal ligament is continuous with the peritoneum over the inferior surface of the diaphragm and anterior surface of the left kidney. The terminal portions of the splenic artery and vein, and, more inferiorly, the tail of the pancreas, lie between the two peritoneal layers of the splenorenal ligament. The tail of the pancreas may be injured during dissection when ligating and dividing the splenic vessels, resulting in bleeding, local pancreatitis and pancreatic fistula formation (Petroianu 2011, Skandalakis et al 1993).

The gastrosplenic ligament is continuous with the phrenicosplenic ligament, the splenic capsule, the gastric serosa and the greater omentum. It contains the short gastric and superior polar arteries, and the left gastroepiploic artery, all of which arise from the splenic artery, and their corresponding veins. During splenectomy or mobilization of the fundus of the stomach, the short gastric vessels must not be ligated too close to the stomach in order to avoid the risk of local gastric necrosis, perforation and their consequences.

The phrenicocolic ligament connects the splenic flexure of the colon to the diaphragm and runs inferior and lateral to the lower pole of the spleen. It is continuous with the peritoneum of the lateral end of the transverse mesocolon at the end of the pancreatic tail, and the splenorenal ligament at the hilum of the spleen (Merchea et al 2012, Skandalakis et al 1993). When the phrenicocolic ligament is being divided, particularly when electrocautery is used, the colon is at risk of injury.

Mobile spleen The length of the peritoneal ligaments associated with the spleen vary; longer ligaments afford the spleen greater mobility, which can stretch its vascular pedicle. This facilitates surgical mobilization but may render the spleen more susceptible to injury from shear forces during trauma.

A floating or wandering spleen is characterized by excessive mobility and migration of the organ outside the left hypochondrium. In such cases, the spleen may undergo torsion or cause bladder or rectal symptoms from local pressure; imaging usually confirms the diagnosis (**Fig. 70.6**). If surgical treatment is necessary, the spleen is freed from local

Operating on a normal-sized spleen is a relatively straightforward procedure for most surgeons. Mobilization of the spleen requires division of the phrenicocolic, gastrosplenic and phrenicosplenic ligaments. Undue traction on the phrenicocolic ligament during mobilization of the splenic flexure may tear the splenic capsule, causing bleeding (Merchea et al 2012). This is less likely if the phrenicocolic ligament is retracted laterally rather than inferiorly and medially (Merchea et al 2012). The anterosuperior border and anterior diaphragmatic surface of the spleen are often adherent to the greater omentum and care must be taken when retracting the latter. The diaphragmatic surface of the spleen is occasionally adherent to the peritoneum on the undersurface of the diaphragm; these adhesions may follow inflammation of the spleen or be congenital in origin (Petroianu 2011, Skandalakis et al 1993).

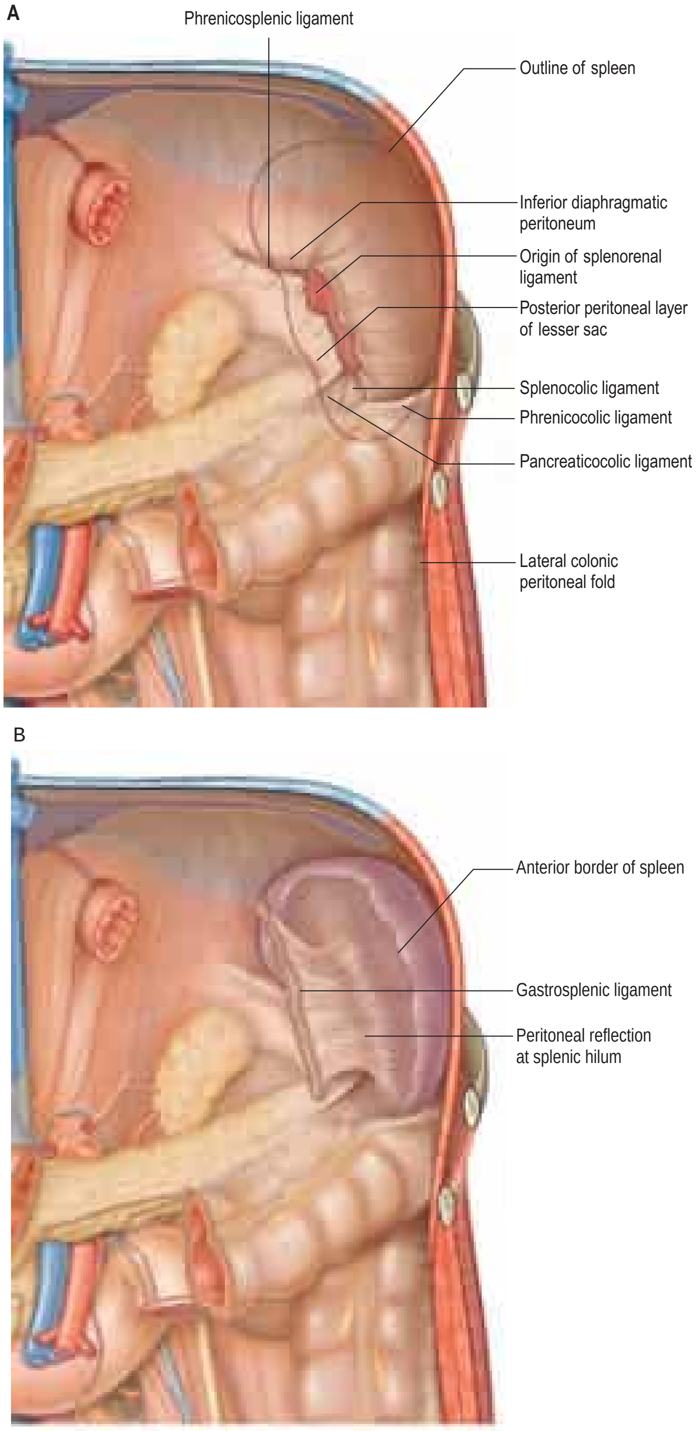

Fig. 70.5 The peritoneal connections of the spleen. **A**, The posterior peritoneal connections, seen as if the spleen has been removed, with the folds fixed in their 'normal' positions. **B**, The anterior peritoneal connections, seen with the spleen in place, as if the stomach and greater omentum have been removed, with the folds fixed in their 'normal' positions. (Adapted from Drake, RL, Vogl, AW, Mitchell, A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

SECTION 8

adhesions and returned to the left hypochondrium, the surgeon shortening and suturing its ligaments to the diaphragm (McFee et al 1995, Petroianu 2011, Üngör et al 2007).

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Arteries

The spleen is supplied by the splenic artery, one of the most tortuous arteries in the body (**Fig. 70.7**). The pathophysiology of the tortuosity of this vessel, which may become more pronounced with advancing age, is not understood, although several theories have been proposed (Borley et al 1995, Cortés and Pellico 1988, García-Porrero and Lemes 1988, Liu et al 1996, Pandey et al 2004, Sylvester et al 1995). Almost always, the splenic artery arises from the coeliac trunk, in common with the left gastric and common hepatic arteries. However, it may originate from the common hepatic artery or the left gastric artery, or rarely directly from the aorta either in isolation or as a splenomesenteric trunk (Cortés and Pellico 1988, García-Porrero and Lemes 1988, Liu et al 1996, Pandey et al 2004, Torres 1998, Trubel 1985).

From its origin, the artery runs a little way inferiorly before turning to the left behind the stomach to run horizontally posterior to the upper border of the body and tail of the pancreas. Multiple loops or even coils of the artery appear above the superior border of the pancreas (McFee et al 1995, Pandey et al 2004). The splenic artery courses anterior to the left kidney and left suprarenal gland, and runs in the splenorenal ligament behind or above the tail of the pancreas. In its course, it gives off numerous branches to the pancreas (dorsal pancreatic, greater pancreatic artery, and arteries to the tail) and, near its termination, it gives off the short gastric arteries and the left gastroepiploic artery (Gürleyik et al 2000, Liu et al 1996, Mikhail et al 1979, Pandey et al 2004, Skandalakis et al 1993, Trubel et al 1985, Trubel et al 1988). Additional branches include a posterior gastric artery in 40% of individuals and small retroperitoneal branches.

The splenic artery varies between 8 and 32 cm in length and its calibre usually exceeds that of the common hepatic and left gastric arteries, ranging from 3 to 12 mm. Splenic artery blood flow is approximately 3 ml/sec/100 g, corresponding to approximately 7% of cardiac output (Cortés and Pellico 1988, García-Porrero and Lemes 1988, Nakamura et al 1989, Pandey et al 2004, Petroianu 2011, Skandalakis et al 1993, Torres 1998, Trubel et al 1985).

The splenic artery usually divides into two, or occasionally three, branches before entering the hilum of the spleen. The superior and inferior branches are sometimes known as superior and inferior polar arteries; as they enter the hilum they divide into four or five segmental arteries that each supply a segment of splenic tissue. There is relatively little arterial collateral circulation between segments, which means that occlusion of a segmental vessel often leads to infarction of part of the spleen (Cortés and Pellico 1988, García-Porrero and Lemes 1988, Gupta et al 1976, Liu et al 1996, Mikhail et al 1979, Pandey et al 2004, Torres 1998, Trubel al 1985, Trubel et al 1988). Segmental arteries divide within the splenic trabeculae and give rise to follicular arterioles, which are surrounded by a thick lymphoid sheath of white pulp. These feed the sinusoids of the red pulp. There is considerable communication between arterioles (García-Porrero and Lemes 1988, Liu et al 1996, Mikhail et al 1979, Skandalakis et al 1993, Sow et al 1991, Trubel et al 1988).

The superior pole of the spleen gains an additional arterial supply, distinct from the splenic hilar vessels, from the short gastric arteries in the gastrosplenic ligament. These vessels connect the superior pole of the spleen to the gastric fundus and preserve viability of this region of the spleen after ligation of the splenic pedicle (García-Porrero and Lemes 1988, Gürleyik et al 2000, Liu et al 1996, Petroianu 2011, Petroianu and Petroianu 1994, Petroianu et al 1989, Skandalakis et al 1993, Torres 1998, Trubel et al 1988).

Veins

Blood from the parenchyma of the spleen is collected by trabecular veins. They join to form segmental veins that drain individual splenic segments. In general, there are no anastomoses between intrasegmental venous tributaries. Segmental veins join to form two (superior and inferior) or three (superior, middle and inferior) 'lobar' veins that emerge from the length of the splenic hilum and unite to form the splenic vein within the splenorenal ligament (see Fig. 70.2; **Fig. 70.8**). Communicating veins may interconnect lobar veins (DeLand 1970, Gupta et al 1980, Liu et al 1996, Par et al 1965, Sow et al 1991).

The splenic vein runs medially below the splenic artery and posterior to the tail and body of the pancreas (see Fig. 69.4 D–G) (Gürleyik et al 2000, Par et al 1965). It crosses the posterior abdominal wall anterior to the left kidney, renal hilum and abdominal aorta, separated from the left sympathetic trunk and left crus of the diaphragm by the left renal vessels, and from the abdominal aorta by the superior mesenteric artery and left renal vein (Gupta et al 1976, Gupta et al 1980, Liu et al 1996, Par et al 1965). It ends posterior to the neck of the pancreas, where it joins the superior mesenteric vein to form the portal vein. Along its course it receives the short gastric veins, left gastroepiploic veins, retroperitoneal veins (Retzius's retroperitoneal venous plexus), pancreatic veins, posterior gastric vein, left gastric vein (occasionally) and the inferior mesenteric vein. Approximately 60% of portal vein blood flow is derived from the gut and the remainder from the spleen and pancreas

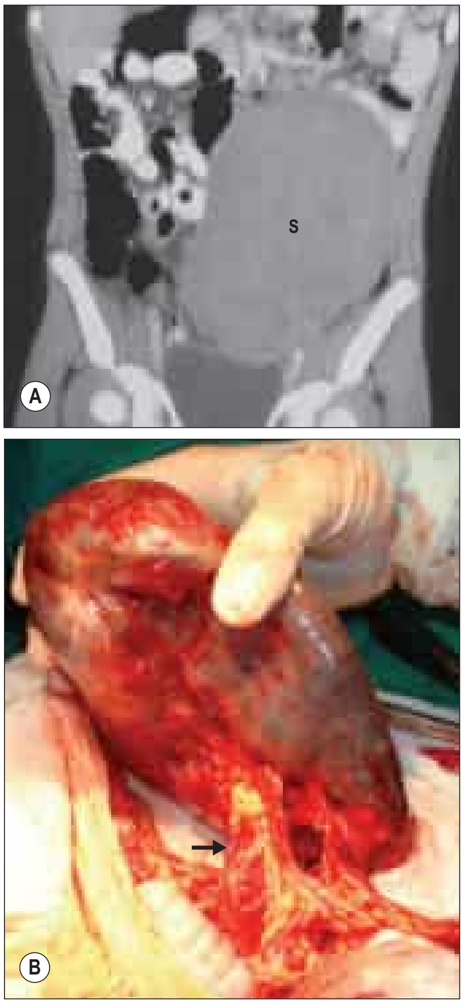

Fig. 70.6 A mobile enlarged spleen located in the left side of the pelvis. **A**, A computed tomography (CT) scan showing the abnormal location of an enlarged spleen (S). **B**, A surgical view of the same spleen, which appears congested due to partial obstruction of its venous drainage at the splenic hilum (arrow). Note the lack of restraining peritoneal ligaments.

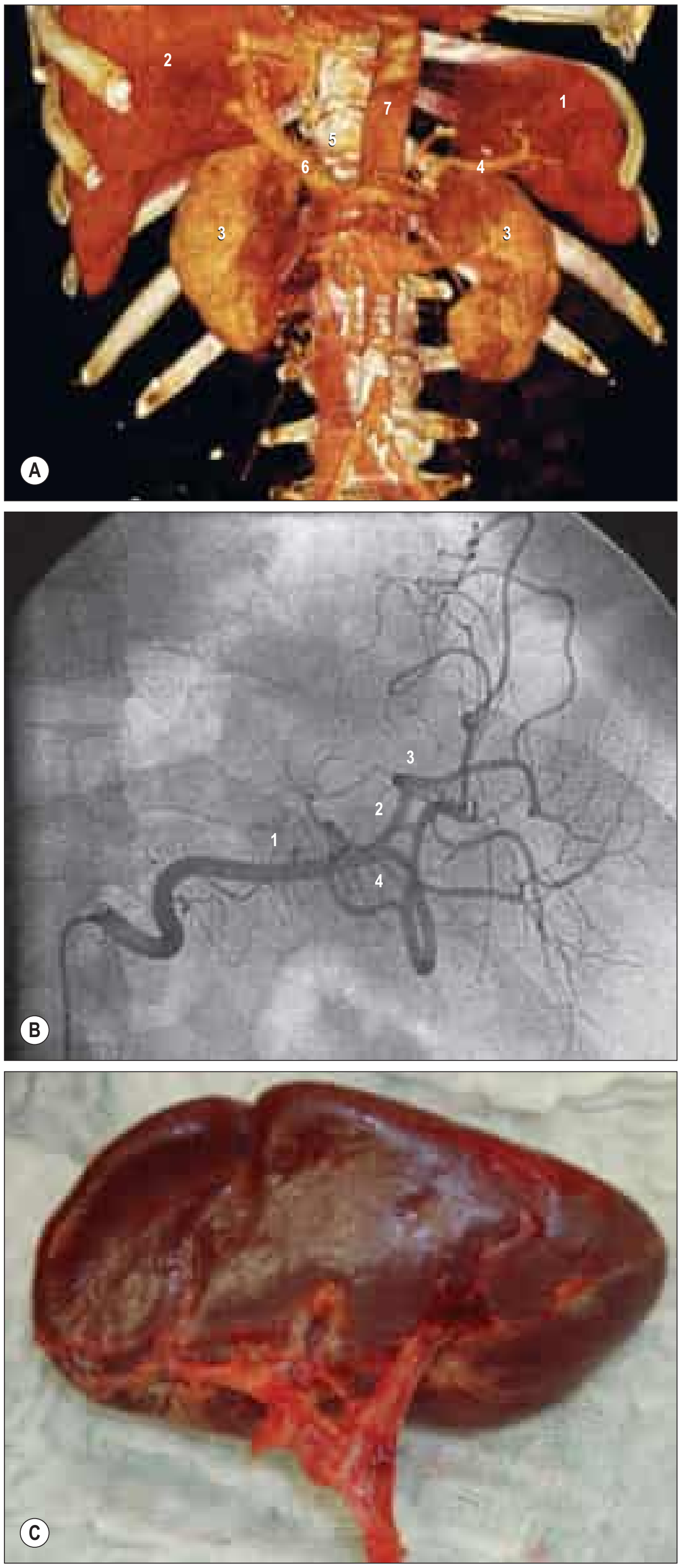

Fig. 70.7 **A**, A three-dimensional reconstructed contrast-enhanced CT scan. Key: 1, spleen; 2, liver; 3, kidneys; 4, splenic vein; 5, superior polar vein; 6, portal vein; 7, aorta. **B**, A splenic digital subtraction arteriogram showing the splenic artery (1) and its branches. Note the primary division of the splenic artery (2) into superior (3) and inferior branches (4). **C**, An excised spleen demonstrating the hilar vessels.

Fig. 70.8 **A**, A splenoportogram. **B**, An axial oblique CT slice showing the portal and splenic veins.

(Dawson et al 1986, Gupta et al 1980, Gürleyik et al 2000, Liu et al 1996, Oguro et al 1993, Par et al 1965).

Obstruction to portal or splenic venous drainage leads to reversed venous flow through the splenic vein and its tributaries, resulting in splenomegaly and widespread varices (Petroianu 1988, Petroianu 1992, Petroianu 2011, Petroianu et al 1989, Re et al 1985, Skandalakis et al 1993).

Lymphatic drainage

Lymphatic drainage begins in the white pulp. Lymphatics travel with the blood vessels towards a lymphatic subcapsular plexus, which drains via larger lymphatic channels to lymph nodes at the splenic hilum and around the tail of the pancreas (**Fig. 70.9**). From here, lymph drains to suprapancreatic, infrapancreatic and omental lymph nodes, and from there to coeliac nodes and the cisterna chyli.

INNERVATION

The spleen is innervated by both components of the autonomic nervous system; the sympathetic supply is dominant. Postganglionic sympathetic nerves from the coeliac plexus and parasympathetic nerves from the vagal trunks travel with the splenic vessels (see Fig. 70.9). Sympathetic fibres innervate arteries at least to the trabecular level and have the potential to influence blood flow within the human spleen (Kudoh et al 1979). Unlike in some animals, the motor innervation of the human splenic capsule is largely vestigial; it contains minimal quantities of smooth muscle and therefore does not contract. In contrast, the capsule and parenchyma are innervated by sensory fibres that convey pain. Mild to moderate splenomegaly is often painless but splenic inflammation from infection, infarction (e.g. sickle cell disease, embolization) or abrupt distension of the capsule (e.g. haematoma) may cause severe pain, intensified by inflammation of the overlying parietal peritoneum. Referred pain from the splenic pulp is poorly localized to the epigastrium (Petroianu 2011).

MICROSTRUCTURE

The spleen contains the largest single mass of lymphoid tissue in the body in direct continuity with the circulation. Splenic parenchyma is divided into two principal regions – white pulp and red pulp – named according to the appearance of the cut surface of a fresh spleen (**Fig. 70.10**).

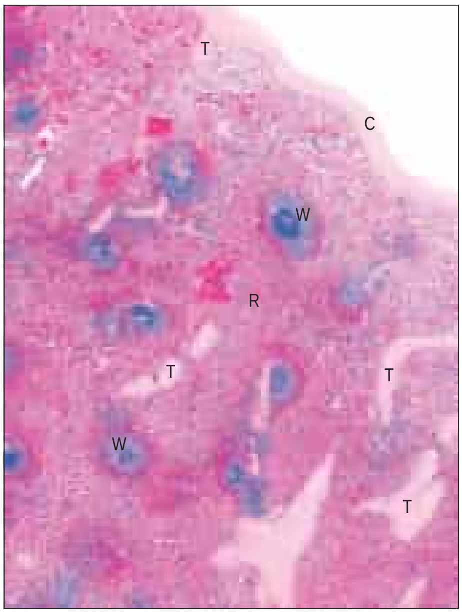

Fig. 70.10 A section through the spleen. White pulp (W) is present as ovoid areas of basophilic tissue. Red pulp (R) lies between white pulp tissue and consists of splenic sinusoids and intervening cellular cords. Part of the capsule (C) is seen top right, from which trabeculae (T) extend into the splenic tissue. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

Fibrous framework

The serosa of the peritoneum covers the entire spleen, except at its hilum and where the peritoneal ligaments are attached. The connective tissue capsule, deep to the serosa, is approximately 1.5 mm thick and contains abundant type I collagen fibres and some elastin fibres. It is composed of an outer and inner lamina in which the directions of the collagen fibres differ, so increasing its strength. Numerous trabeculae extend from the capsule into the substance of the spleen, where they branch to form a connective tissue scaffold (Faller 1985, Scothorne 1985). The largest trabeculae enter at the hilum and divide into branches in the splenic pulp, providing conduits for the splenic vessels and nerves (see Fig. 70.10; **Figs 70.11–70.12**). Within the spleen, these branches become continuous with a delicate network of type III (reticular) collagen fibres that pervades both red and white pulps and is maintained by fibroblasts within its interstices (Gupta et al 1976, Krieken and Velde 1988, Scothorne 1985).

White pulp

In an adult, white pulp accounts for between 5% and 20% of the splenic tissue. Branches of the splenic artery radiate out into the parenchyma of the spleen from the hilum, ramifying within trabeculae. In their terminal few millimetres, their connective tissue adventitia is replaced by a sheath of T lymphocytes, the peri-arteriolar lymphatic sheath (PALS). This is expanded in places by aggregations of B lymphocytes, lymphoid follicles measuring 0.25–1 mm in diameter and visible to the naked eye on the freshly cut surface of the spleen as white semi-opaque dots, in contrast to the surrounding deep reddish purple of the red pulp. Follicles are usually situated near the terminal branches of arterioles and typically protrude to one side of a vessel, which consequently appears eccentrically placed within the follicle. Arterioles branch laterally within follicles to form a series of parallel terminal arterioles, penicillar arterioles (Krieken and Velde 1988, Scothorne 1985; see Figs 70.10–70.12).

Like peri-arteriolar sheaths, follicles are centres of lymphocyte aggregation and proliferation. After antigenic stimulation, they become sites of intensive B-cell proliferation, developing germinal centres similar to those found in lymph nodes; antigen presentation by follicular dendritic cells is involved in this process. Germinal centres regress when the stimulus abates. Follicles tend to atrophy with advancing age and may be absent in the very elderly (Krieken and Velde 1988, Petroianu 2011, Scothorne 1985; see Figs 70.10–70.12).

Red pulp

The red pulp constitutes up to 90% of the total splenic volume and is a unique filtration device that enables the spleen to clear particulate material from the blood as it perfuses the organ (Krieken and Velde 1988, Petroianu 2011, Scothorne 1985; see Fig. 70.11). It contains large numbers of venous sinusoids that ultimately drain into tributaries of the splenic vein. The sinusoids are separated from each other by a fibrocellular network of small bundles of collagen fibres, the reticulum, numerous reticular fibroblasts and splenic macrophages. Seen in two dimensions, these intersinusoidal regions appear as strips of tissue, the splenic cords (see Figs 70.11–70.12), whereas in reality they form a three-dimensional continuum around the venous spaces (Chen 1978, Krieken and Velde 1988, Scothorne 1985, Weiss 1983).

Venous sinusoids are elongated ovoid vessels, approximately 50 µm in diameter, supported externally by circumferential and longitudinal reticular fibres (Krieken and Velde 1988, Scothorne 1985, Weiss 1983), and lined by a characteristic, discontinuous endothelium that is unique to the spleen. The long, narrow endothelial cells are aligned with the long axis of the sinusoid; they are also called stave cells because they resemble the planks in a barrel (see Fig. 70.12). Stave cells are attached to their neighbours at intervals along their length by short stretches of intercellular junctions that alternate with intercellular slits that allow blood cells to squeeze into the lumen of the sinusoid from the surrounding splenic cords. A discontinuous basal lamina is present on the abluminal aspect of the sinus.

Large, stellate fibroblasts – reticular cells – lie around the sinusoids. They synthesize the matrix components of the reticulum, including collagen and proteoglycans, and their cytoplasmic processes help to compartmentalize the reticular space. Blood from the open ends of the capillaries that originate from penicillar arterioles percolates through the reticular spaces within the splenic cords. Macrophages in the spaces remove blood-borne particulate material, including ageing and damaged erythrocytes. If the number of damaged erythrocytes increases, the

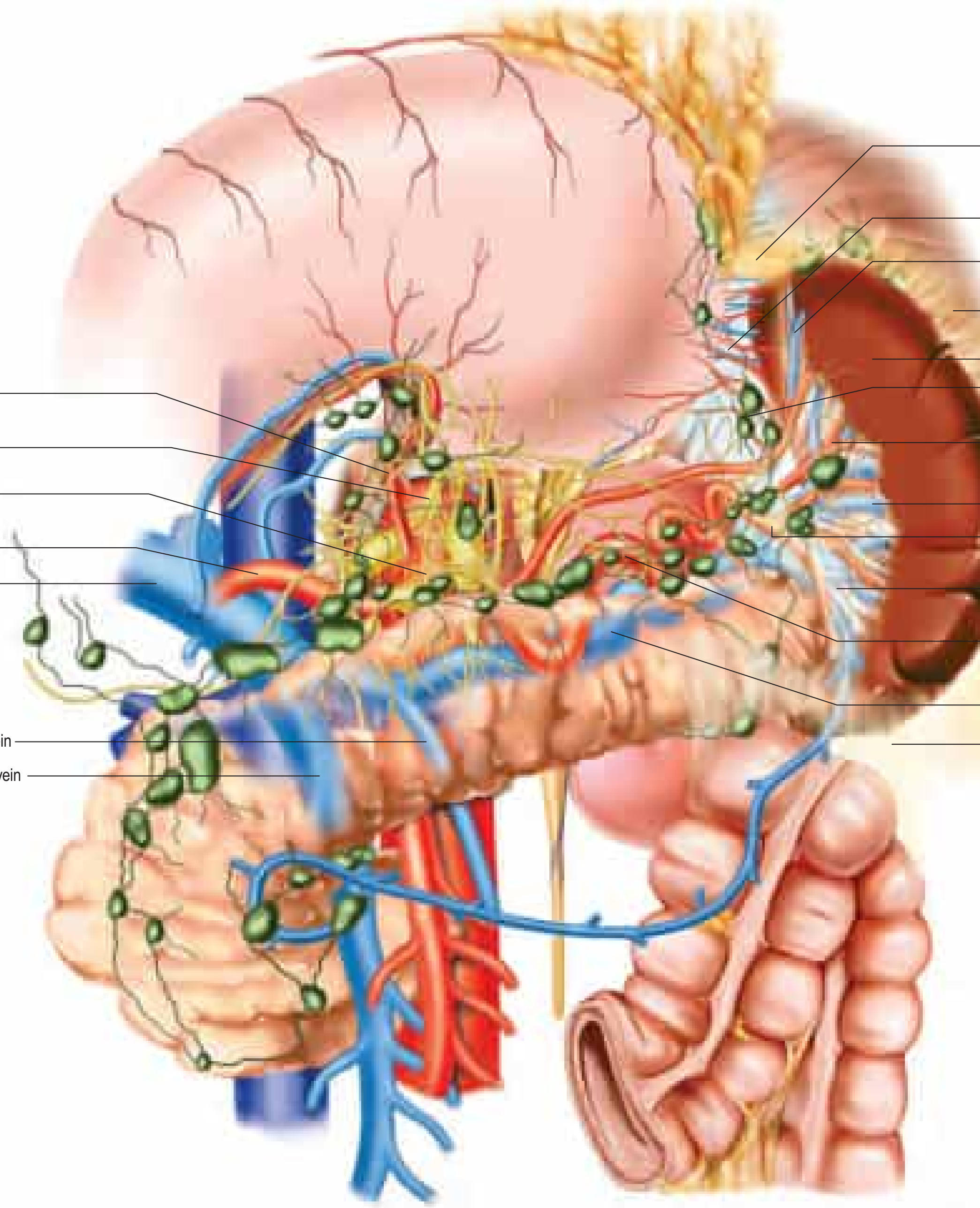

Fig. 70.9 The lymph nodes related to the spleen. The pancreas has been rendered partially transparent to visualize the major blood vessels lying posteriorly. The greater curvature of the stomach has been reflected superiorly to expose its posterior surface and the peritoneal lining of the posterior wall of the lesser sac removed. (Courtesy of Dr Andy Petroianu and Dr Iriam Starling.)

In malignant neoplasms of the greater curvature of the stomach, distal pancreas or distal transverse colon, it may be necessary to remove the spleen and tail of the pancreas to achieve a radical resection. In leukaemias and lymphomas, lymphadenopathy of splenic hilar nodes may be found along with splenomegaly during staging investigations (Petroianu 2011).

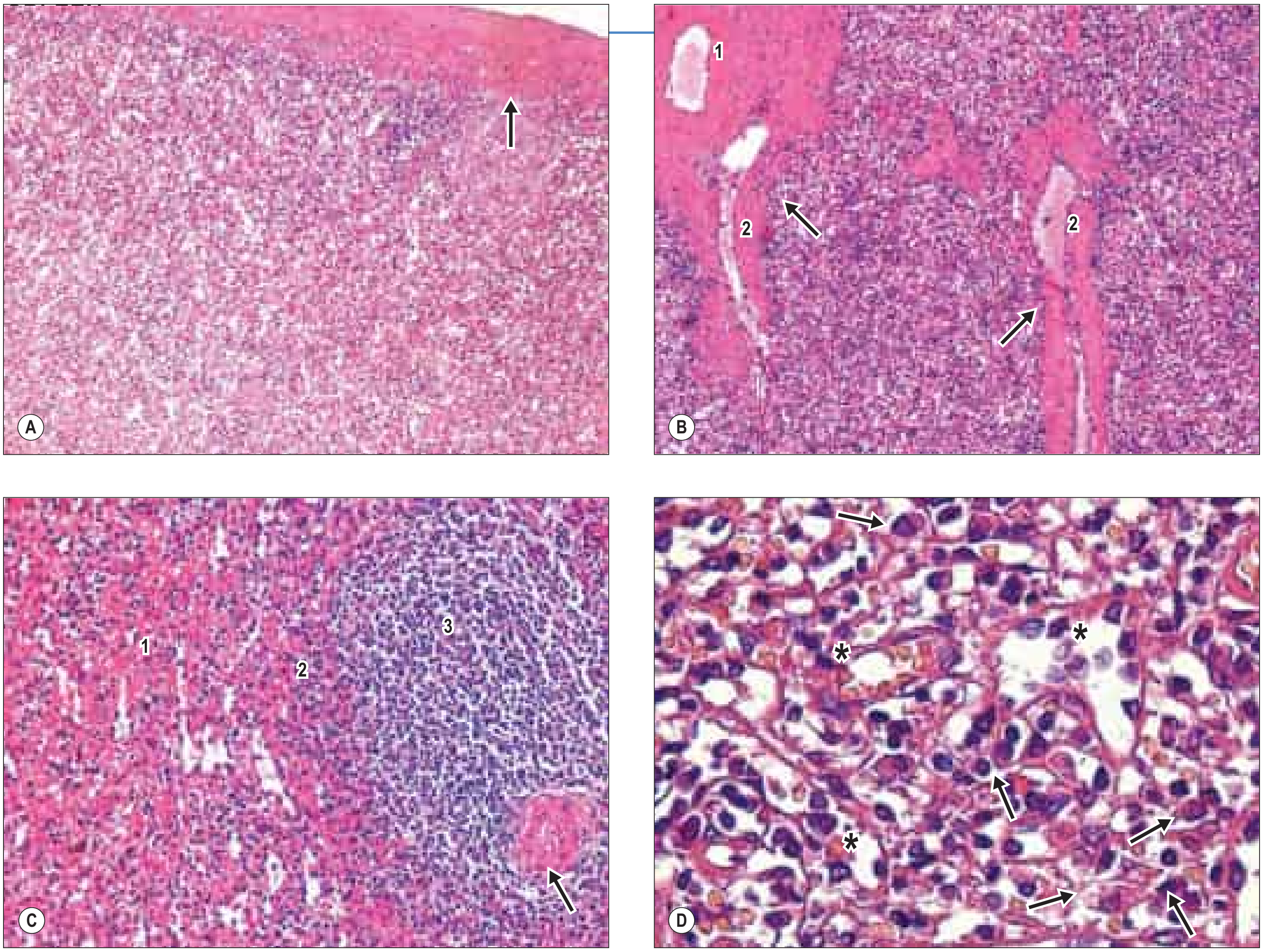

Fig. 70.11 The microstructure of the spleen (haematoxylin and eosin staining). **A**, Red pulp covered by the splenic capsule (arrow). **B**, Trabeculae (arrows) inside the red pulp. Trabecular arterioles (1) and venules (2), containing erythrocytes, are also shown. **C**, Red pulp (1), marginal or perifollicular zone (2), and white pulp (3) containing a follicular arteriole surrounded by a peri-arteriolar lymphatic sheath (arrow). **D**, Red pulp containing several sinusoid capillaries (*) with discontinuous endothelial cells and surrounded by macrophages containing phagocytosed particles within their cytoplasm (arrows).

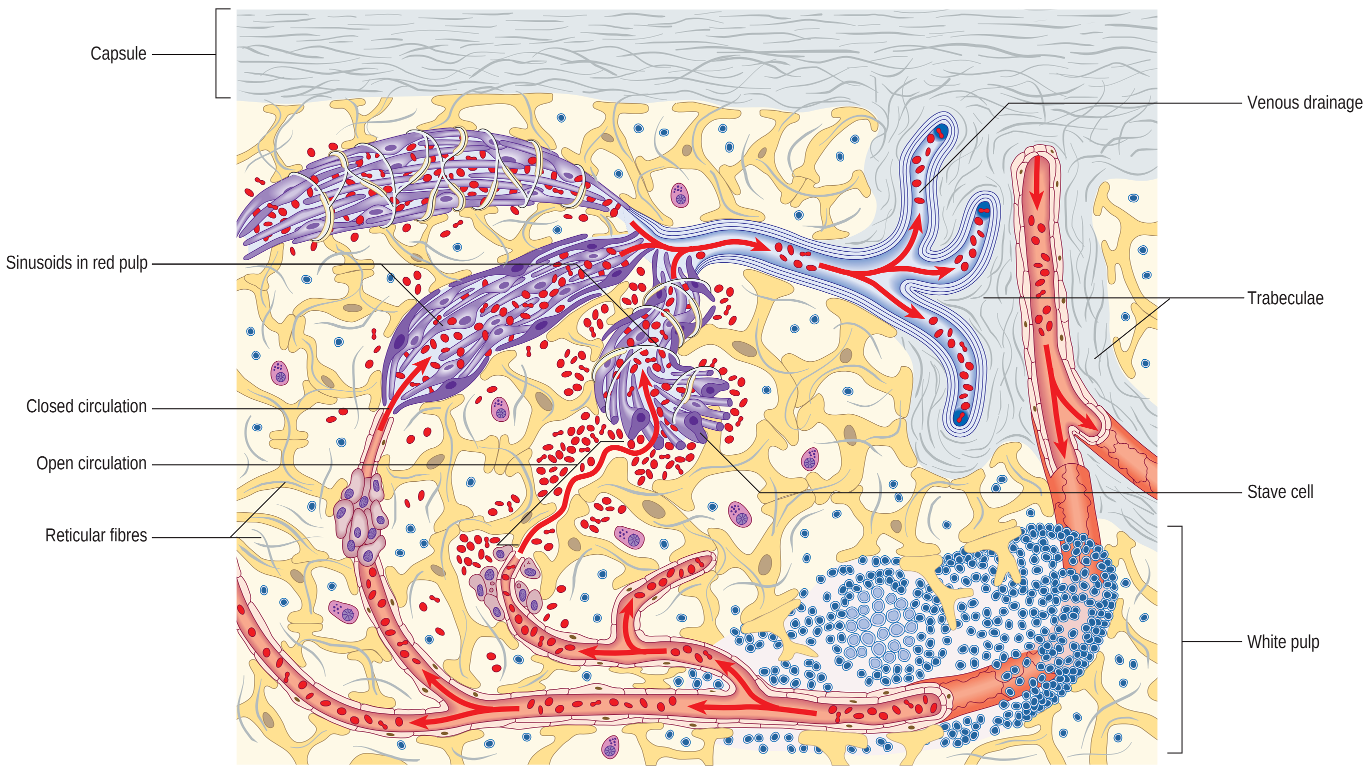

Fig. 70.12 The main features of splenic structure, not to scale. The capsule, trabeculae, reticular fibres and cells, the perivascular lymphoid sheaths and follicles (white pulp), and the cellular cords and venous sinusoids of the red pulp are shown. The 'open' and 'closed' theories of splenic circulation are illustrated; it is likely that most of the circulation is of the open form. The venous sinusoids are lined by specialized endothelial 'stave' cells; the intercellular gaps of these have been over-emphasized for clarity.

reticular cells proliferate and the red pulp expands, causing the spleen to enlarge (Kudoh et al 1979, Scothorne 1985).

Perifollicular zone

The perifollicular (marginal) zone lies at the interface between the white and red pulp; it is the site where blood is delivered into the red pulp and where many lymphocytes leave the circulation to migrate to their respective T- and B-lymphocyte aggregations in the white pulp. These lymphocytes are more loosely arranged than they are in the white pulp, and are held in a dense network of reticular fibres and cells. The arterioles leaving the white pulp are also surrounded by a small aggregation of macrophages: the peri-arteriolar macrophage sheath (Krieken and Velde 1988, Petroianu 2011, Scothorne 1985; see Fig. 70.11C).

Splenic microcirculation

Segmental splenic arteries ramify in the trabeculae (trabecular arteries) before tapering to become arterioles that pass through the white pulp and give off penicillar arterioles (García-Porrero and Lemes 1988, Gupta et al 1976, Scothorne 1985). These exit the white pulp and traverse the perifollicular marginal zone before entering the red pulp. Evidence supports a dominantly 'open' circulation in humans, in which blood empties into and percolates slowly through the reticular tissue of the splenic cords before re-entering the vascular lumen through slits in the walls of the venous sinusoids (Cavalli et al 1982, Chen 1978; see Fig. 70.12). From the venous sinusoids, blood collects in venules that unite to form small veins that run within the trabeculae and drain into segmental splenic veins (Cavalli et al 1982, Chen 1978, Petroianu 2011, Scothorne 1985). Some blood probably takes an alternative, 'closed' route and enters the venous sinusoids directly from arterioles and capillaries in the marginal zone. Either way, the blood is exposed to macrophages and the filtering mechanism of the spleen responsible for removing abnormal and effete cells, blood cell particles, bacteria, parasites and foreign antigens.

IMAGING

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SPLENOMEGALY

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SPLENIC TRAUMA

The spleen is one of the most frequently damaged organs in blunt abdominal trauma (**Table 70.2**, **Fig. 70.15**). It is particularly prone to injury during rapid deceleration or compression, and its peritoneal ligaments render it vulnerable to shearing injuries of the parenchyma and splenic vessels. Fractures of the lower left ribs may be associated with blunt injuries (Buntain et al 1988). Traction on peritoneal ligaments or adhesions may lead to inadvertent capsular tears during surgical procedures such as splenic flexure mobilization. There is insufficient evidence to know whether enlarged spleens are at a significantly greater risk of trauma (Skandalakis et al 1993), but this is frequently postulated.

SPLENECTOMY

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Bonus e-book images

Fig. 70.1 A supernumerary spleen located in the greater omentum.

Fig. 70.6 A mobile enlarged spleen located in the left side of the pelvis. **A**, A computed tomography (CT) scan showing the abnormal location of an enlarged spleen. **B**, A surgical view of the same spleen, which appears congested due to partial obstruction of its venous drainage at the splenic hilum.

Fig. 70.7 A, A three-dimensional reconstructed contrast-enhanced CT scan. **B**, A splenic digital subtraction arteriogram showing the splenic artery and its branches. **C**, An excised spleen demonstrating the hilar vessels.

Fig. 70.8 A, A splenoportogram. **B**, An axial oblique CT slice showing the portal and splenic veins.

Fig. 70.9 The lymph nodes related to the spleen.

Fig. 70.13 Splenic imaging. **A**, An ultrasound scan with colour Doppler. **B**, A magnetic resonance scan. **C**, A radioisotope scan (^{99m}Tc sulphur colloid).

Fig. 70.14 Examples of different pathologies causing splenomegaly. **A**, Portal hypertension. **B**, Gaucher's disease. **C**, Myelofibrosis. **D**, Thalassaemia. **E**, Sarcoma. **F**, Lymphoma.

Fig. 70.15 Splenic trauma. **A**, A CT scan showing multiple splenic tears. **B**, Suture repair of a ruptured spleen. **C**, Complete splenic rupture. **D**, Subtotal splenectomy after treatment of the patient illustrated in C.

Fig. 70.16 Partial splenectomy and autotransplantation. **A**, Partial splenectomy preserving the superior pole supplied by the short gastric vessels. **B**, Partial splenectomy preserving the inferior pole supplied by the left gastroepiploic vessels. **C**, A splenic segment removed after total splenectomy and sliced into 1–2 cm cubes. **D**, The same slices in C sutured on to the greater omentum.

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A description of the tortuosity of the splenic artery, demonstrating its importance in anatomical and surgical approaches.

Imaging demonstrates the gross morphology and vascular supply of the spleen, and can also be used to assess splenic function. Imaging modalities include ultrasound (US), computed tomography (CT), magnetic resonance imaging (MRI), arteriography, splenoportography and scintigraphy (Buntain et al 1988, Merran et al 2007, Nakamura et al 1989, Xu et al 2009; see Figs 70.7–70.8; **Fig. 70.13**). Cross-sectional imaging is particularly useful for assessing tumours and infiltrations of the spleen. Percutaneous drainage of splenic and perisplenic fluid collections is routinely performed under image guidance, and percutaneous splenic biopsy is being increasingly utilized in diagnosis. Arteriography is essential for selective embolization and the insertion of an arterial stent to treat splenic artery aneurysm. Scintigraphy can be used to investigate splenic phagocytic function by quantifying the uptake of radioactive labelled agents (e.g. ^{99m}technetium sulphur colloid or heat-damaged erythrocytes) (Hermann and Winkel 1977, Merran et al 2007, Petroianu 2011, Xu et al 2009).

The numerous causes of splenomegaly may broadly be classified into haematological (e.g. lymphoproliferative, myeloproliferative, erythrocyte disorders); vascular (e.g. portal hypertension); infectious (e.g. viral, bacterial, other); autoimmune; and infiltrative (e.g. storage disorders and tumours) (Cavalli et al 1982, Petroianu 1988, Petroianu 2011, Re et al 1985; **Fig. 70.14**).

Contrast-enhanced CT imaging is particularly useful in the assessment of splenic trauma. The majority of splenic injuries from blunt abdominal trauma are successfully treated by conservative management. Extensive burst injuries with persistent bleeding or major injuries to the hilar vessels may require splenectomy but various repair techniques are available. With hilar injuries, a subtotal splenectomy, preserving the short gastric vessels and superior pole only, is one option. Minor intraoperative capsular tears may be controlled by the topical application of haemostatic agents and sutures, although direct suture repair is challenging because the splenic pulp is fragile. Arteriography and embolization are additional techniques that can be used to control bleeding or treat vascular lesions (Petroianu 2011).

Most surgical procedures performed on the spleen should start with ligation of the splenic artery at the superior border of the pancreatic tail. This can be difficult in patients with portal hypertension, splenic artery aneurysm, retroperitoneal fibrosis, regional lymphadenopathy or inflammatory conditions. After splenic artery ligation, the spleen becomes softer because most of its contained blood returns to the circulation. Despite the interruption of splenic artery inflow, the superior pole of the spleen continues to be perfused by the short gastric arteries, and the inferior pole may be supplied retrogradely via the left gastroepiploic artery. Ligation and division of the splenic vein should be the last step before removal of the spleen to maximize autotransfusion of blood from the spleen (Liu et al 1996, Petroianu 2011, Petroianu and Petroianu 1994, Petroianu et al 1989, Skandalakis et al 1993, Sow et al 1991).

In partial or subtotal splenectomy, the vessels that supply the part of the spleen to be removed are ligated and divided close to the organ. The devascularized region turns dark blue, demarcating it from the residual viable splenic tissue; the organ is then divided in this plane (Christo and DiDio 1997, Gürleyik et al 2000, Liu et al 1996, Petroianu and Petroianu 1994, Sow et al 1991). Wedge excision leaves two flaps of splenic capsule that can be sutured or stapled to cover the remaining raw parenchyma, after suture of any visible bleeding vessels (Petroianu 2011, Petroianu 1993, Petroianu et al 1989, Petroianu 1988; **Fig. 70.16**).

If total splenectomy is required to treat splenic trauma, autotransplantation of splenic tissue into the greater omentum or mesentery of the bowel can preserve useful splenic function (see Fig. 70.16). Approximately 20% of the original splenic mass needs to be transplanted and venous drainage of this tissue must be to the portal venous system to maintain the metabolic and immunological functions of the splenic tissue (Oguro et al 1993, Petroianu 2011).

Total splenectomy has numerous potential consequences. Intraoperatively, bleeding or damage to the pancreas, stomach or colon may occur. Early postoperative complications include left-sided basal pulmonary atelectasis and pleural effusion, subphrenic abscess, transient leukocytosis and thrombocytosis with the risk of thrombotic complications. In the longer term, there is an increased risk of infection (especially from encapsulated bacteria), which may be fatal (Petroianu 2011).

Conservative management of splenic trauma and partial splenectomy now enable splenic function to be preserved in situations where splenectomy would once have been routinely performed. Furthermore, most surgical procedures on the spleen can now be performed laparoscopically, which readily affords inspection of the spleen from all perspectives (Poulin and Thibaut 1993, Sow et al 1991).

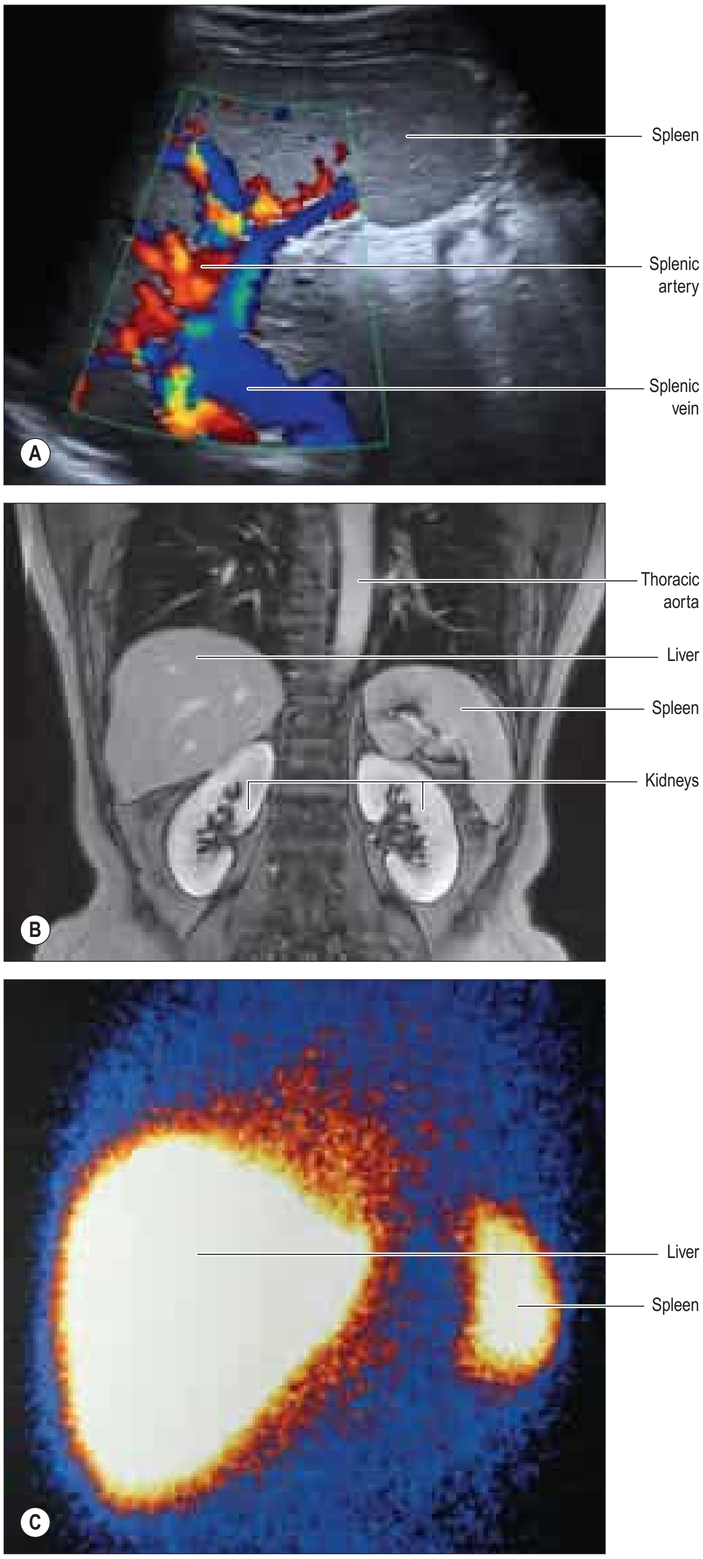

Fig. 70.13 Splenic imaging. **A**, An ultrasound scan with colour Doppler. **B**, A magnetic resonance scan. **C**, A radioisotope scan (^{99m}Tc sulphur colloid). Contrast-enhanced CT imaging, digital subtraction angiography and a splenoportogram are shown in Figures 70.7–70.8.

Table 70.2 **Buntain's classification of splenic injury**

Grade	Description of injury
Grade I	Subcapsular haematoma or localized capsular disruption without significant injury to the parenchyma
Grade II	Single or multiple capsular and parenchymal disruptions that do not involve the hilar vessels
Grade III	Deep fractures in the hilum involving the vessels
Grade IV	Shattered or fragmented spleen or hilar avulsion

(Reproduced from Buntain WL, Gould HR, Maull KI. Predictability of splenic salvage by computed tomography. J Trauma. 1988; 28:24–34.)

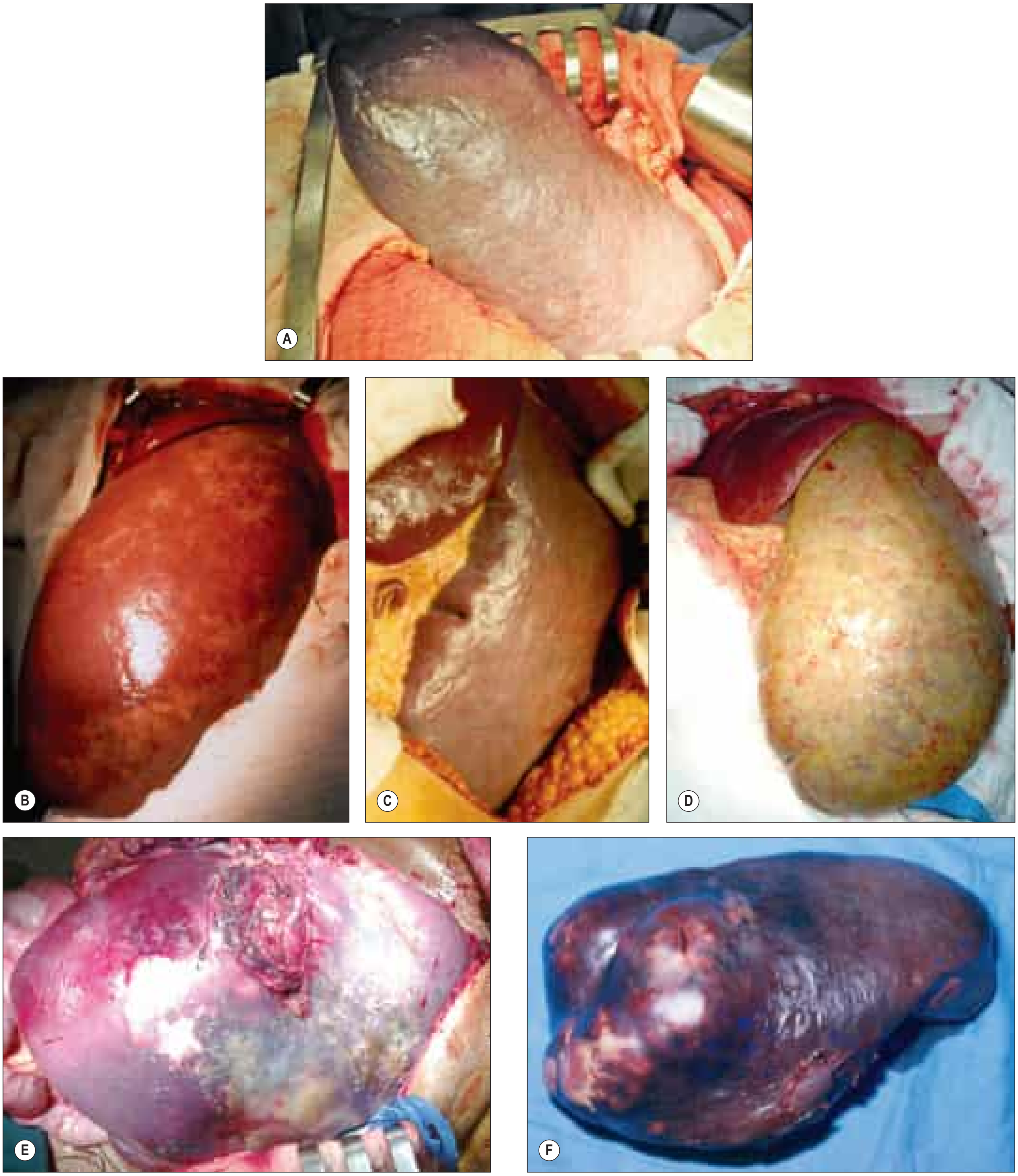

Fig. 70.14 Examples of different pathologies causing splenomegaly. **A**, Portal hypertension. **B**, Gaucher's disease. **C**, Myelofibrosis. **D**, Thalassaemia. **E**, Sarcoma. **F**, Lymphoma.

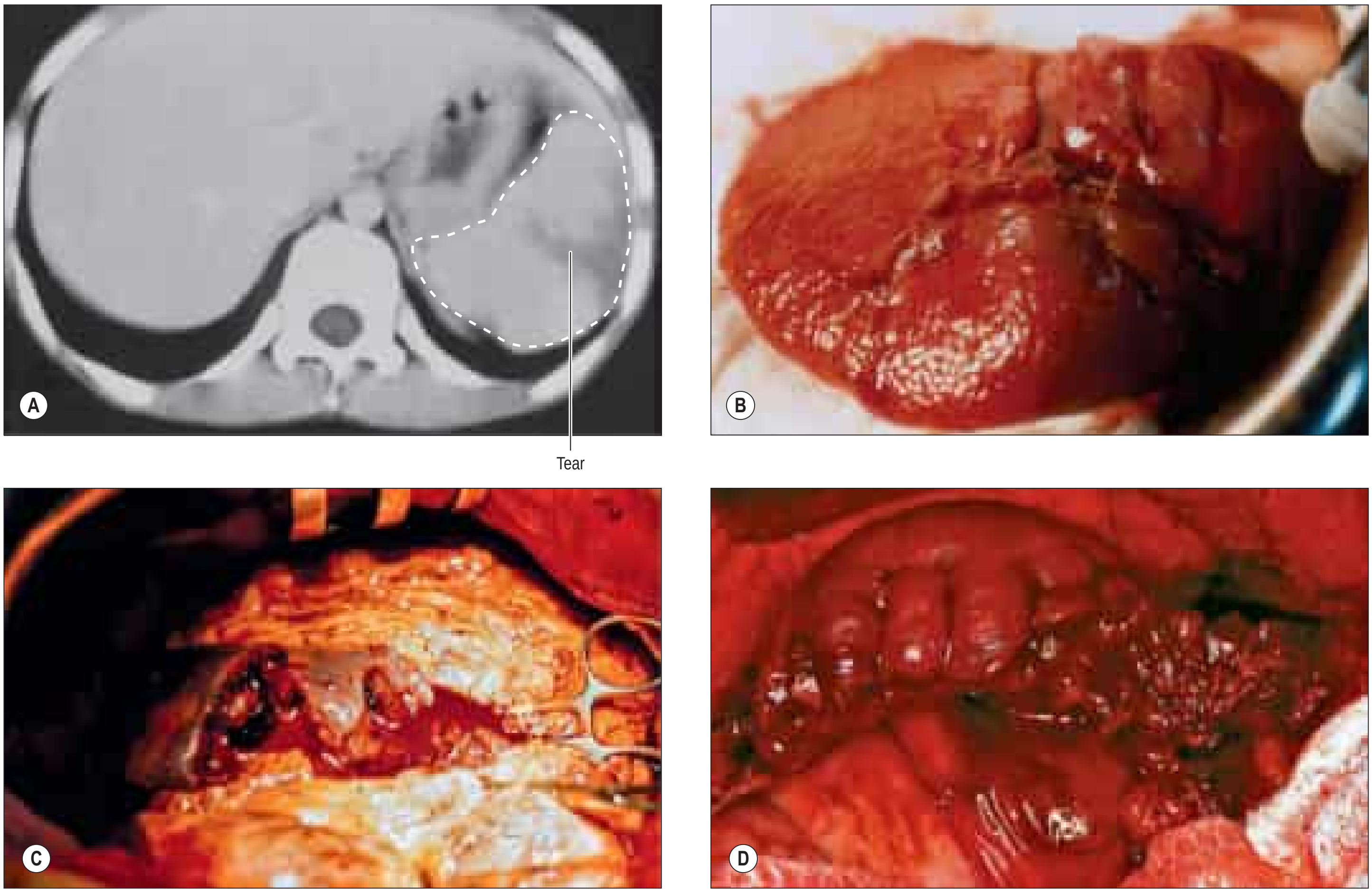

Fig. 70.15 Splenic trauma. **A**, A CT scan showing multiple splenic tears. **B**, Suture repair of a ruptured spleen. **C**, Complete splenic rupture. The splenic hilum has been clamped to control bleeding. **D**, Subtotal splenectomy after treatment of the patient illustrated in C.

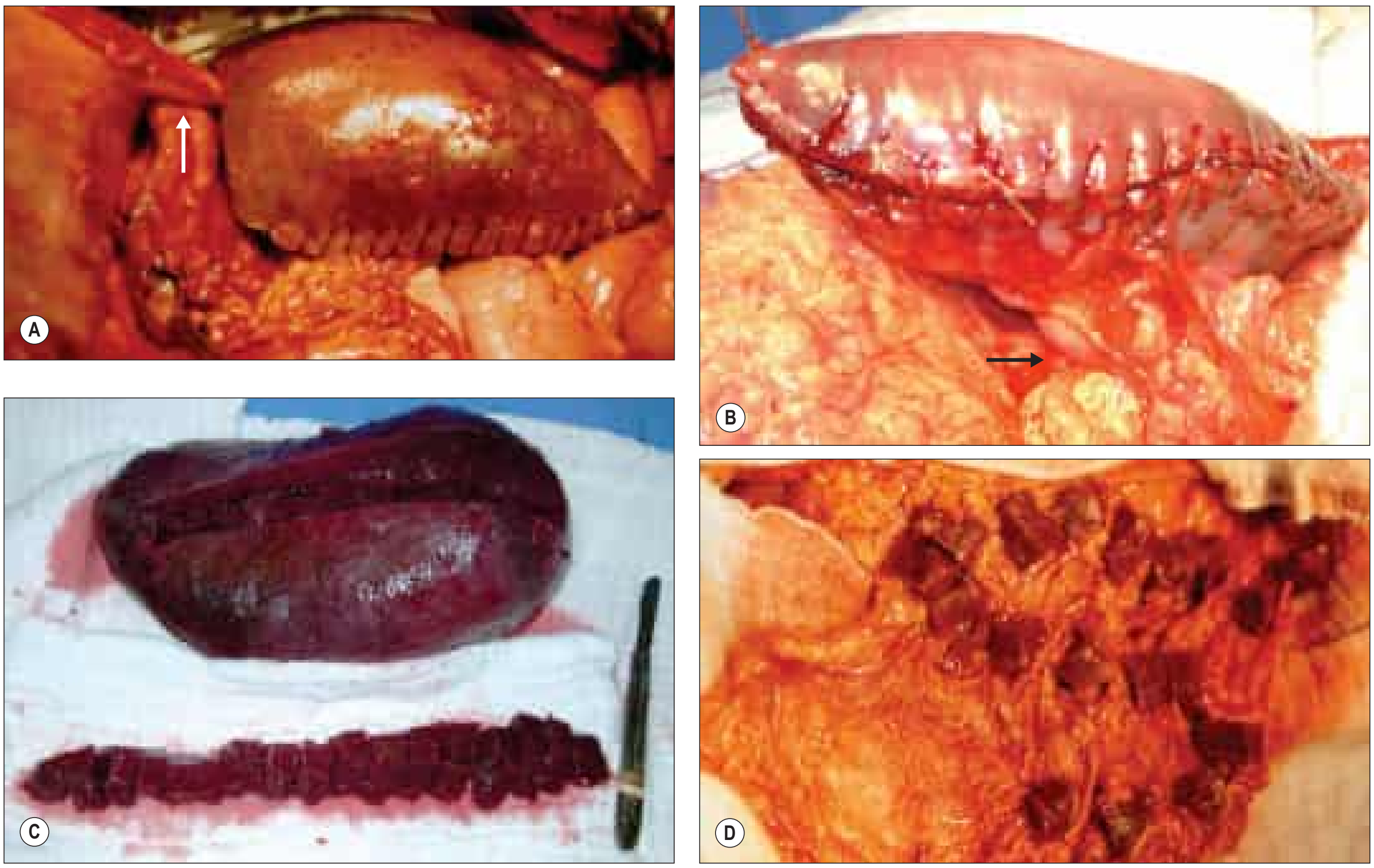

Fig. 70.16 Partial splenectomy and autotransplantation. **A**, Partial splenectomy preserving the superior pole supplied by the short gastric ('splenogastric') vessels (arrow). **B**, Partial splenectomy preserving the inferior pole supplied by the left gastroepiploic vessels (arrow). **C**, A splenic segment removed after total splenectomy and sliced into 1–2 cm cubes. **D**, The same slices in C sutured on to the greater omentum.

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